

7 (a) stress or $\sigma = F/A$ C1

$$\text{max. tension} = \text{UTS} \times A = 4.5 \times 10^8 \times 15 \times 10^{-6} = 6800 \text{ (6750) N}$$

A1 [2]

(b) $\rho = m/V$ C1

$$\begin{aligned} \text{weight} &= mg = \rho Vg = \rho ALg \\ 6750 &= 7.8 \times 10^3 \times 15 \times 10^{-6} \times L \times 9.81 \end{aligned}$$

C1

$$L = 5.9 \text{ (5.88)} \times 10^3 \text{ m}$$

A1

or

$$\begin{aligned} \text{maximum mass} &= 6750/9.81 = 688 \text{ kg} && \text{(C1)} \\ \text{mass per unit length} &= \rho A = 0.117 \text{ kg m}^{-1} && \text{(C1)} \\ L &= 688/0.117 = 5.9 \times 10^3 \text{ m} && \text{(A1)} \end{aligned}$$

or

$$\begin{aligned} \text{maximum mass} &= 6750/9.81 = 688 \text{ kg} && \text{(C1)} \\ \text{volume} &= m/\rho = 0.0882 \text{ m}^3 = LA && \text{(C1)} \\ L &= 0.0882/15 \times 10^{-6} = 5.9 \times 10^3 \text{ m} && \text{(A1)} \end{aligned}$$

[3]